Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

1.44 g of titanium (At. mass $= 48$) reacted with excess of O_2 and produce xg of non stoichiometric compound $Ti_{1.44}O$. The value of x is:

(A) 2

(B) 1.77

(C) 1.44

(D) none of these

Q.2

An organic compound whose empirical and molecular formula are same, contains 20% carbon, 6.7% hydrogen, 46.7% nitrogen and the rest oxygen. On heating it yields ammonia, leaving a solid residue. The solid residue gives a violet colour with dilute solution of alkaline copper sulphate. The organic compound is

(A) NH_2COONH_4

(B) $HCOONH_4$

(C) NH_2NHCHO

(D) NH_2CONH_2

Q.3

The mass of N_2F_4 produced by the reaction of 2.0 g of NH_3 and 8.0 g of F_2 is 3.56 g. What is the per cent yield?

$2NH_3 + 5F_2 \longrightarrow N_2F_4 + 6HF$

(A) 79.0

(B) 71.2

(C) 84.6

(D) None of these

Q.4

The impure 6 g of $NaCl$ is dissolved in water and then treated with excess of silver nitrate solution. The mass of precipitate of silver chloride is found to be 14 g. The $\%$ purity of $NaCl$ solution would be:

(A) 95%

@IITJEE_Advanced

(B) 85%

(C) 75%

(D) 65%

Q.5

A gaseous compound of nitrogen and hydrogen contains 12.5% (by mass) of hydrogen. The density of the compound relative to hydrogen is 16.
The molecular formula of the compound is:

(A) NH_2

(B) N_3H

(C) NH_3

(D) N_2H_4

Q.6

What is the empirical formula of vanadium oxide, if $2.74\ g$ of the metal oxide contains $1.53\ g$ of metal?

(A) V_2O_3

(B) VO

(C) V_2O_5

(D) V_2O_7

Q.7

$3.92\ g$ of ferrous ammonium sulphate crystals are dissolved in $100\ mL$ of water. $20\ mL$ of this solution requires $18\ mL$ of potassium permaganate during titration for complete oxidation. The weight of $KMnO_4$ present in one litre of the solution of

(A) $3.47\ g$

(B) $12.38\ g$

(C) $1.23\ g$

(D) $34.76\ g$

Q.8

Specific volume of cylindrical virus particle is $6.02 \times 10^{-2} cc/g$. whose radius and length are $7\mathring{A} \& 10\mathring{A}$ respectively.
If $N_A = 6.02 \times 10^{23}\ mol^{-1}$, find molecular weight of virus

(A) $3.08 \times 10^3\ kg/mol$

@IITJEE_Advanced

(B) $3.08 \times 10^4 \text{ kg/mol}$

(C) $1.54 \times 10^4 \text{ kg/mol}$

(D) 15.4 kg/mol

Q.9

10 moles SO_2 and 15 moles O_2 were allowed to react over a suitable catalyst. 8 moles of SO_3 were formed. The remaining moles of SO_2 and O_2 respectively are -

(A) 2 moles, 11 moles

(B) 2 moles, 8 moles

(C) 4 moles, 5 moles

(D) 8 moles, 2 moles

Q.10

On reduction with hydrogen 3.6 g of an oxide of metal leaves 3.2 g of metallic residue. If the atomic mass of metal is 64 , the formula of metal oxide is

(A) M_2O_3

(B) M_2O

(C) MO

(D) MO_2

Q.11

What is the mass of precipitate formed when 50 mL of 16.9% solution of $AgNO_3$ is mixed with 50 mL of $5.8\% NaCl$ solution? $(Ag = 107.8,\ N = 14, O = 16, Na = 23, Cl = 35.5)$

(A) 28 g

(B) 3.5 g

(C) 7 g

(D) 14 g

Q.12

A transition metal M forms a volatile chloride which has a vapour density of 94.8. If it contains 74.75% of chlorine the formula of the metal chloride will be

(A) MCl_3

@IITJEE_Advanced

(B) MCl_2

(C) MCl_4

(D) MCl_5

Q.13

The amount of $BaSO_4$ formed upon mixing $100ml$ of $20.8\% BaCl_2$ solution with 50 ml of $9.8\% H_2SO_4$ solution will be:
(Given that molecular weight of $Ba = 137, Cl = 35.5,\ S = 32, H = 1$ and $O = 16\ g/mol$)

(A) $23.3\ g$

(B) $11.65\ g$

(C) $30.6\ g$

(D) $33.2\ g$

Q.14

A gas mixture of 3 litres of propane (C_3H_8) and butane (C_4H_{10}) on complete combustion at $25^\circ C$ produced 10 litre CO_2. Find out the composition of gas mixture (Propane: Butane)

(A) $2:1$

(B) $1:2$

(C) $1.5:1.5$

(D) $0.5:2.5$

Q.15

What volume of hydrogen gas, at $273\ K$ and $1\ atm.$ pressure will be consumed in obtaining $21.6\ g$ of elemental boron (atomic mass $= 10.8$) from the reduction of boron trichloride by hydrogen ?

(A) 67.2

(B) 67.1

(C) 67.3

(D) 67.0

Q.16

9 moles of "D" and 14 moles of E are allowed to react in a closed vessel according to given reactions.
Calculate number of moles of B formed in the end of reaction, if 4 moles of G are present in reaction vessel.
(Percentage yield of reaction

@IITJEE_Advanced

is mentioned in the reaction)

Step-1 $3\mathrm{D} + 4\mathrm{E} \xrightarrow{80\%} 5\mathrm{C} + \mathrm{A}$

Step-2 $3\mathrm{C} + 5\mathrm{G} \xrightarrow{50\%} 6\ \mathrm{B} + \mathrm{F}$

(A) 2.4

(B) 30

(C) 4.8

(D) 1

Q.17

A mixture of $\mathrm{NH_4NO_3}$ and $(\mathrm{NH_4})_2\ \mathrm{HPO_4}$ contain 30.40% mass per cent of nitrogen. What is the mass ratio of the two components in the mixture?

(A) $2:1$

(B) $1:2$

(C) $3:4$

(D) $4:1$

Q.18

What is the volume of $\mathrm{CO_2}$ liberated (in litres) at 1 atmosphere and $0^\circ\mathrm{C}$ when $10\ \mathrm{g}$ of 100% pure calcium carbonate is treated with excess dilute sulphuric acid? (Atomic mass $\mathrm{Ca}:40, \mathrm{C}:12, \mathrm{O}:16$)

(A) 0.224

(B) 2.24

(C) 22.4

(D) 224

Q.19

$20\ \mathrm{g}$ of $\mathrm{CaCO_3}$ on heating gave $8.8\ \mathrm{g}$ of $\mathrm{CO_2}$ and $11.2\ \mathrm{g}$ of CaO. This is in accordance with

(A) The law of conservation of mass.

(B) The law of constant composition.

(C) The law of reciprocal proportion.

(D) None of these

Q.20

A gaseous mixture contains oxygen and nitrogen in the ratio of $1:4$ by weight. Therefore, the ratio of their number of molecules is

@IITJEE_Advanced

(A) $1:4$

(B) $1:8$

(C) $7:32$

(D) $3:16$

Answers & Solutions

Q.1 Answer:

1.77

Solution:

$$\mathrm{Ti} + \mathrm{O_2} \longrightarrow \mathrm{Ti_{1.44}O}$$

$\frac{1.44}{48}$ **mole** $\frac{\mathrm{x}}{48(1.44)+16}$ **mole**

$$\therefore \frac{1.44}{48} = \frac{1.44\mathrm{x}}{48(1.44)+16}$$

$$\mathrm{x} = 1.77\ \mathrm{g}$$

Q.2 Answer:

$\mathrm{NH_2CONH_2}$

Solution:

\begin{array}{|c|c|c|c|}

**\hline Element & Percentage & Atomic mass & Relative no. of atoms & Simplest ratio **

**\hline C & 20 \% & 12 & 1.66 & 1 **

**\hline H & 6.7 \% & 1 & 6.7 & 4 **

**\hline N & 46.7 \% & 14 & 3.33 & 2 **

**\hline O & 26.6 \% & 16 & 1.66 & 1 **

\hline

\end{array}

Empirical formula = Molecular formula

$= \mathrm{CH_4\ N_2O}$ **or** $\mathrm{NH_2CONH_2}$

$$\mathrm{H_2NCONH_2 + H_2NCONH_2} \xrightarrow{\Delta} \mathrm{H_2NCONHCONH_2 + NH_3}$$

biuret,

When an aqueous solution of biuret is treated with dilute sodium hydroxide and a drop of copper sulphate, a violet colour is produced. This test is known as biuret test, and is characteristic of compounds having the group $-\mathrm{CONH}-$

Q.3 Answer:

None of these

Solution:

@IITJEE_Advanced

$\underset{34\ \text{g}}{2\text{NH}(\text{g})} + 5\ \underset{190\ \text{g}}{\text{F}_2} \rightarrow \underset{104\ \text{g}}{\text{N}_2\ \text{F}_4} + 6\text{HF}$

Amount of $\text{N}_2\ \text{F}_4$ formed by $2\ \text{g}\text{NH}_3$ $= \frac{104}{34} \times 2 = 6.12$

Amount of $\text{N}_2\ \text{F}_4$ formed by $8\ \text{g}\ \text{F}_2$ $= \frac{104}{190} \times 8 = 4.38$

$\text{N}_2\ \text{F}_4$ will be limiting and actual amount of product is $3.56\ \text{g}$

$\therefore 96\ \text{yield} = \frac{\text{Actual amount of product}}{\text{Calculated amount of product}} \times 100$

$= \frac{3.56}{4.38} \times 100 = 81.2896$

Q.4 Answer:

95%

Solution:

The reaction that takes place is

$\text{NaCl} + \text{AgNO}_3 \longrightarrow \text{AgCl} \downarrow + \text{NaNO}_3$

$\therefore 143.5\ \text{g}$ **of** AgCl **is produced from** $58.5\ \text{g}\text{NaCl}$

$\therefore 14\ \text{g}$ **of** AgCl **will be produced from**

$\frac{58.5 \times 14}{143.5} = 5.70\ \text{g}\text{NaCl}$

This is the amount of NaCl **in common salt:**

$\%$ **purity** $= \frac{5.70}{6} \times 100 = 95\%$

Q.5 Answer:

N_2H_4

Solution:

In an unknown compounds containing N **and** H **given** $\%$ **of** $\text{H} = 12.5\%$

$\therefore \quad \%$ **of** $\text{N} = 100 - 12.5 = 87.5\%$

\begin{array}{|c|c|c|c|}

**\hline Element & Percentage & Atomic ratio & Simple ratio **

**\hline \mathrm{H} & 12.5 \% & \frac{12.5}{1}=12.5 & \frac{12.5}{6.25}=2 **

**\hline \mathrm{N} & 87.5 & \frac{87.5}{14}=6.25 & \frac{6.25}{6.25}=1 **

\hline

\end{array}

$2\times$ **vapour density** $=$ **Mol.** $\text{wt} = \text{molwt.}$

$= 16 \times 2 = 32$

Molecular formula $= n\times$ **empirical formula**

mass $n = \frac{32}{16} = 2$

$\therefore$ **Molecular formula of the compound will be** $= (\text{NH}_2)_2 = \text{N}_2\text{H}_4$

Q.6 Answer:

V_2O_5

Solution:

@IITJEE_Advanced

Mass of oxide = Mass of metal + Mass of oxygen $2.74 = 1.53 + \mathrm{W}_{\text{Oxygen}} \Rightarrow \mathrm{W}_{\text{Oxygen}} = 1.21\ \mathrm{g}$

Moles of $\mathrm{V} = \frac{1.53}{51} = 0.03$

Moles of $\mathrm{O} = \frac{1.21}{16} = 0.075$

$\mathrm{V}_{0.03}\mathrm{O}_{0.075}$

$\mathrm{VO}_{2,5} \Rightarrow \mathrm{V}_2\mathrm{O}_5$

Q.7 Answer:

$3.47\ \mathrm{g}$

Solution:

Normality of ferrous amm. sulphate

$= \frac{3.92 \times 1000}{392 \times 100} = 0.1$(**Eq. wt of FAS is 392**)

$\mathrm{N}_1\ \mathrm{V}_1 = \mathrm{N}_2\ \mathrm{V}_2$

$20 \times 0.1 = 18 \times \mathrm{N}_2 \quad \mathrm{N}_2 = 0.111$

g ev. of $\mathrm{KMnO}_4 = 31.6\ \mathrm{g}$

$0.111\ \mathrm{g}$ **ev. of** $\mathrm{KMnO}_4 = 31.6 \times 0.111 = 3.5\ \mathrm{g}.$

Q.8 Answer:

$15.4\ \mathrm{kg/mol}$

Solution:

Specific volume (volume of $1\ \mathrm{g}$ **) of cylindrical virus particle** $= 6.02 \times 10^{-2}\mathrm{cc/g}$

Radius of virus $(\mathrm{r}) = 7\mathring{A} = 7 \times 10^{-8}\ \mathrm{cm}$

Length of virus $= 10 \times 10^{-8}\ \mathrm{cm}$

Volume of virus $=$

$\pi r^2 1 = \frac{22}{7} \times \left(7 \times 10^{-8}\right)^2 \times 10 \times 10^{-8}$

$= 154 \times 10^{-23}\mathrm{cc}$

Wt. of one virus particle $= \frac{\text{volume}}{\text{specific volume}}$

$\therefore$ **Mol. wt. of virus = Wt. of** N_A **particle**

$= \frac{154 \times 10^{-23}}{6.02 \times 10^{-2}} \times 6.02 \times 10^{23} = 15400\ \mathrm{g/mol}$

$= 15.4\ \mathrm{kg/mol}$

Q.9 Answer:

2 moles, 11 moles

Solution:

$$\begin{array}{ccccc} 2\mathrm{SO}_2 & + & \mathrm{O}_2 & \longrightarrow & 2\mathrm{SO}_3 \\ 10 & & 15 & & 0 \\ 10-2x & & 15-x & & 2x \end{array}$$

$\because 2x = 8 \quad x = 4$

@IITJEE_Advanced

Hence, remaining, $SO_2 = 10 - 8 = 2$ **moles,**
$O_2 = 15 - 4 = 11$ **moles**

Q.10 Answer:

M_2O

Solution:

Mass of oxygen which gets displaced from metal oxide $= 0.4$ g
Now, 0.4 g **of oxygen combines with metal** $= 3.2$ g
8 g **of oxygen combines with metal** $\frac{3.2}{0.4} \times 8 = 64$ g
GEW of metal $= 64$ g
Valency of metal $= \frac{\text{GAM}}{\text{GEW}} = \frac{64\text{ g}}{64\text{ g}} = 1$
Hence, formula of oxide is M_2O.

Q.11 Answer:

7 g

Solution:

50 mL **of** 16.9% **solution of** $AgNO_3$
$\left(\frac{16.9}{100} \times 50\right) = 8.45$ g **of** $AgNO_3$
$n_{mole} = \frac{8.45\text{ g}}{(107.8+14+16\times3)\text{g/mol}} = \left(\frac{8.45\text{ g}}{169.8\text{ g/mol}}\right)$
$= 0.0497$moles
50ml **of** 5.8% **solution of** NaCl **contain** $NaCl = \left(\frac{5.8}{100} \times 50\right) = 2.9$ g

$$n_{NaCl} = \frac{2.9\text{ g}}{(23+35.5)\text{g/mol}} = 0.0495\text{moles}$$

$AgNO_3 + NaCl \rightarrow AgCl\downarrow + Na\odot + Cl\ominus$

1 mole $\quad$ **1 mole** $\quad$ **1 mole** $\therefore$ 0.049 **mole** 0.049 **mole** $\quad$ 0.049 **mole of** $AgCl$

$n = \frac{w}{M}$
$w = (n_{AgC}) \times$ **Molecular Mass**
$= (0.049) \times (107.8 + 35.5)$
$= 7.02$ g

Q.12 Answer:

MCl_4

Solution:

74.75% **of chlorine means** 74.75 g **chlorine is present in** 100 g **of metal chloride. Weight of metal**
$= 100$ g $- 74.75$ g $= 25.25$ g

@IITJEE_Advanced

$$= \frac{\text{weight of metal}}{\text{weight of chlorine}} \times 35.5$$

$$= \frac{25.25}{74.75} \times 35.5 = 12$$

Equivalent weight

$$\text{Valency of metal} = \frac{2 \times \text{V.D.}}{\text{Equivalent wt. of metal} + 35.5}$$

$$= \frac{2 \times 94.8}{12 + 35.5} = 4$$

$\therefore$ **Formula of compound** $= MCl_4$

Q.13 Answer:

11.65 g

Solution:

100ml **of** $20.8\%BaCl_2$ **solution** $= 20.8$ g$BaCl_2 = 0.1$ mol.

50ml **of** $9.8\%H_2SO_4$ **solution** $= 4.9$ g$H_2SO_4 = 0.05$ mol.

The reaction is as follows:

$BaCl_2 + H_2SO_4 \leftrightharpoons BaSO_4 + 2HCl$

Since, H_2SO_4 **is the limiting reagent, only** 0.05 **moles of** $BaSO_4$ **will form.**

Hence, $0.05 \times 233 = 11.65$ g **of** $BaSO_4$ **is formed.**

Q.14 Answer:

$2 : 1$

Solution:

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

a $3a$

$C_4H_{10} + \frac{13}{2}O_2 \rightarrow 4CO_2 + 5H_2O$

$(3 - a)$

$4(3 - a)$

But, $3a + 4(3 - a) = 10$

$a = 2$ **(Propane) and** $3 - 2 = 1$ **(Butane**

Q.15 Answer:

67.2

Solution:

$2BCl_3 + 3H_2 \rightarrow 2\,B + 6HCl$

or $BCl_3 + \frac{3}{2}H_2 \rightarrow B + 3HCl$

Now, since 10.8gm **boron requires hydrogen** $= \frac{3}{2} \times 22.4$ L **at N.T.P**

hence 21.6gm **boron requires hydrogen** $\frac{3}{2} \times \frac{22.4}{10.8} \times 21.6 = 67.2$ L **at N.T.P.**

@IITJEE_Advanced

Q.16 Answer:

2.4

Solution:

$3\mathrm{D} + 4\mathrm{E} \xrightarrow{80\%} 5\mathrm{C} + \mathrm{A}$

9 mole 14 mole $\frac{5}{3} \times 9 \times 0.8 = 12$ **mole**

$\underset{12 \text{ mole}}{3C} + \underset{4 \text{ mole}}{5\mathrm{G}} \xrightarrow{50\%} 6\ \mathrm{B} + \mathrm{F}$

Limiting Reagent is G

$\therefore$ **Moles of** B **formed** $= \frac{6}{5} \times 4 \times 0.5 = 2.4$

Q.17 Answer:

$2 : 1$

Solution:

Let wt. of $\mathrm{NH_4NO_3}$ **and** $(\mathrm{NH_4})_2\,\mathrm{HPO_4}$ **are** x

and y gram respectively

$\frac{\frac{x}{80} \times 2 \times 14 + \frac{y}{132} \times 2 \times 14}{x+y} \times 100 = 30.4$

$\Rightarrow x : y = 2 : 1$

Q.18 Answer:

2.24

Solution:

$\mathrm{CaCO_3} + \mathrm{H_2SO_4} \rightarrow \mathrm{CaSO_4} + \mathrm{H_2O} + \mathrm{CO_2}$

1 mol 100 g 22.4 litre

1 mol 10 g 2.24 litre

Q.19 Answer:

The law of conservation of mass.

Solution:

$\mathrm{CaCO_3} \rightarrow \mathrm{CaO} + \mathrm{CO_2}$

20g 8.8g 11.2g

mass of reactant $=$ **mass of products** $= 20$ g **Hence the law of conservation of mass is obeyed.**

Q.20 Answer:

$7 : 32$

Solution:

Let the weight of oxygen $= x$

The weight of nitrogen $= 4x$

Number of molecules of oxygen $= \frac{x}{32} \times N_0$

@IITJEE_Advanced

Number of molecules of nitrogen $= \frac{4x}{28} \times N_0$

$\frac{\text{number of molecules of oxygen}}{\text{number of molecules of nitrogen}} = \frac{x \times 28}{32 \times 4x} = \frac{7}{32}$

@IITJEE_Advanced